Mika Ismayilli

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EDUCATION

Boston University, Questrom School of Business Boston, MA
M.S. in Business Analytics. *Aug 2025 – Jan 2027 (expected)*
Coursework: Intro to Python Programming, Fundamentals of Data Analysis, Business Analytics Toolbox, Introduction to Data Analytics (QST BA 780), Supervised Machine Learning, Competing with Analytics, Storytelling with Data, Teaming, Analytics for Business (QST BA 775), Unsupervised and Unstructured Machine Learning (QST BA 820), Data Ethics: Analytics in Social Context (QST BA 840), Pricing Strategy and Tactics (QST MK 864), Data Driven Marketing Decision (QST MK 852)

Suffolk University Boston, MA
Bachelor of Science in Psychology, Minor in Business Analytics. *Sep 2023 – May 2025*
Dean's List, Magna Cum Laude.
Coursework: Introduction to Business Analytics (ISOM 130), Data and Decisions Analysis (ISOM 201), Big Data, Business Intelligence and Analytics (ISOM 230), Applied Statistics and Predictive Analytics (ISOM 330), Business Foundations (SBS 101), Business Finance (FIN 200), Introduction to Artificial Intelligence and Machine Learning (CAS 204), Research Methods and Design (PSYCH 216), Cognitive Neuroscience (PSYCH 312)

EXPERIENCE

Admissions Ambassador, M.S. in Business Analytics Oct 2025 – Present
Boston University, Questrom School of Business *Boston, MA*
- Selected as an Admissions Ambassador for Boston University's M.S. in Business Analytics program, representing the program to prospective students and running outreach to applicants weighing the degree against other analytics programs
- Serve as a current-student point of contact for prospective students, giving first-hand perspective on the curriculum, the pre-program analytics bootcamp, and the LAUNCH experience so applicants can judge the fit for themselves

Data Analysis Intern Jun 2026 – Aug 2026
Outfox Energy (Redfish Systems) *Leicester, UK (Remote)*
- Automated recurring reporting for commercial and operations teams by building refreshable Power BI dashboards with DAX measures and Power Query transformations, eliminating manual ad-hoc spreadsheet reporting
- Analyzed direct-debit levels against actual consumption to flag over- and under-set customer accounts, supporting credit-balance and arrears monitoring in line with Ofgem expectations
- Queried a 572-table SQL Server reporting database and cleaned, validated, and explored large operational datasets in Python (pandas) to surface consumption and payment anomalies for the analytics team

Sales Management Intern Jun 2025 – Aug 2025
Daikin Azerbaijan (NEP Engineering) *Baku, Azerbaijan*
- Developed and executed sales strategies for technical and engineering service solutions, driving new client acquisitions across energy, infrastructure, and industrial sectors
- Managed 15 major client accounts and built partnerships with government bodies, private firms, and international contractors, increasing average contract values by 30%
- Led a sales team of five, improving pipeline visibility and forecasting accuracy while collaborating with engineering teams to ensure compliance with client and regulatory requirements

Business Support Coordinator Sep 2024 – Jun 2025
Sustainable Business Network of Massachusetts *Cambridge, MA*
- Supported a network of 50+ Black-owned businesses through technical assistance and direct connections to funding opportunities and strategic partners, including Cambridge Savings Bank and the City of Cambridge
- Managed the Equity Fund application cycle, reviewing 40+ applications annually and collaborating with 10-15 grantees to develop individualized success plans and coordinate technical assistance providers
- Produced and distributed targeted digital content including newsletters, social media posts, and campaigns, contributing to a 30% increase in engagement for member businesses
- Cultivated and maintained partnerships across 5+ affiliate organizations, enhancing access to resources and services that address the operational and strategic needs of Black entrepreneurs

Business Support Intern Jun 2024 – Sep 2024
Sustainable Business Network of Massachusetts *Cambridge, MA*

- Evaluated 20+ businesses across the City of Cambridge for grant eligibility, ensuring compliance with program guidelines and equity-focused criteria

Research Intern Jan 2024 – May 2024
Cambridge Local First *Cambridge, MA*

- Supported CLF's Resilient Economies Internship, contributing to the development of sustainable community initiatives aimed at fostering economic resilience
- Conducted comprehensive research to identify and recommend effective strategies for developing impactful community campaigns, enhancing local engagement by 25%
- Measured and evaluated campaign impact on community awareness and engagement with local institutions, providing actionable insights to improve outreach effectiveness

PROJECTS

BeatTheStreet: Earnings-Surprise Prediction 2026
Independent project, deployed live. Solo build

- Built and deployed a full-stack machine learning application predicting quarterly earnings surprises (beat, meet, or miss) for US public companies, trained on 104,938 earnings observations spanning 2005 to 2024, with a FastAPI backend, a React and TypeScript frontend, and Docker packaging
- Reached 60.9% test accuracy on a strict 2022 to 2024 holdout, a 13.1 percentage point lift over the always-beat baseline of 47.8%, using a LightGBM model tuned on a time-ordered 2005-2019 train, 2020-2021 validation, and 2022-2024 test split with no shuffling so that lookahead bias was prevented
- Engineered features from four WRDS source groups, I/B/E/S analyst estimates and revisions, Compustat fundamentals and accruals, CRSP price momentum and volatility, and FRED macro indicators, applying 1st and 99th percentile winsorization and median imputation before fitting
- Added a VADER news-sentiment layer scoring headlines around each earnings announcement, and surfaced a SHAP waterfall chart on every prediction so a user can see which features pushed the model toward its answer

Job Agent: AI Job-Search Assistant for Claude Code Aug 2026 – Oct 2026
Independent project, open source. Extended and rebuilt a friend's tool; solo work. github.com/MikaIsmayilov/Job-Agent-Public

- Extended a friend's personal job-search agent into an open-source tool that non-technical users set up by chatting with Claude Code, adding a guided onboarding that converts any resume into structured facts and search targets, and 123 automated tests built around a fictional candidate so it works for whoever installs it
- Rebuilt fit scoring (35 to 95) around how many of a posting's requirement lines the resume covers, weighted by skill rarity across postings, and added work-authorization screening that flags citizenship and no-sponsorship requirements with the quoted text
- Rebuilt the application dashboard on Next.js, Cloudflare Workers and D1 behind Cloudflare Access, with a response-rate funnel, follow-up reminders and automated Gmail reply tracking that routes ambiguous replies to human review
- Audited the automated reply tracker against a real inbox, found it was missing most employer decisions, and fixed the classification so ambiguous replies are routed to review instead of guessed

Scarcity Messaging Experiment: Does "Only 3 left!" Change Purchase Intent? 2026
Boston University, BA830. Modeling and regression lead (team project)

- Designed and ran a randomized online experiment in Qualtrics testing whether scarcity messaging causally increases purchase intent, perceived urgency, and anticipated regret, randomizing respondents between a standard product listing and one carrying a scarcity cue
- Analyzed a respondent-by-product long panel of 180 observations from 60 respondents in Python with pyfixest, using covariate-adjusted OLS, product fixed effects with respondent-clustered standard errors, and interaction terms to estimate heterogeneous treatment effects
- Measured a statistically significant increase in purchase intent and perceived urgency ($p < 0.05$ in the adjusted model), found larger effects among respondents scoring higher on impulsivity, and showed that respondent-clustered standard errors were consequential because naive standard errors understated uncertainty by roughly 30%

Airbnb Boston Price and Booking Prediction 2026
Coursework project. Classification track lead

- Built an end-to-end supervised machine learning pipeline on 3,500+ Boston listings with scikit-learn, training a regression model for nightly price and a classification model for 30-day booking likelihood

Formula One Performance Analytics: Multi-Decade Race Data Analysis 2026
Boston University, QSTBA 775 (Analytics for Business). Project manager

- Analyzed three decades of Formula One data with Python and SQL to examine driver performance, constructor dominance, lap-time variability, and circuit effects, and built interactive Tableau dashboards visualizing qualifying versus final results and circuit-level trends

Alone TV Show: Unsupervised Learning and Survival Pattern Analysis 2026
Boston University, QSTBA 820 (Unsupervised and Unstructured Machine Learning). Item-category analysis
- Applied unsupervised learning techniques including K-Means clustering, PCA, and hierarchical clustering in Python to segment contestants across multiple seasons of History Channel's Alone, identifying behavioral and demographic patterns tied to survival outcomes

Semiconductor Trade Resilience: Assessing Global Supply Chain Vulnerabilities 2025
Boston University, QSTBA 780 (Introduction to Data Analytics). Data cleaning and pipeline lead
- Ran a multi-source analysis of global semiconductor trade combining UN COMTRADE flows, wafer fault rates, and the SOXX index, using Python with Pandas, NumPy, NetworkX, GeoPandas, and Seaborn to map supply-chain dependencies and identify fragile nodes

Aurelian: AI Personal Stylist and Wardrobe Intelligence Platform 2026
Independent project, in progress (aurelianfits.lovable.app). Solo build
- Built an AI-powered styling web app that converts user-uploaded clothing into a structured wardrobe database covering category, color, pattern, material, seasonality, and formality, and generates occasion-specific outfits through natural language chat

BU SHIP Chatbot: RAG vs. LLM-Only Coverage Q&A 2026
Boston University, BA840 (Data Ethics: Analytics in Social Context). Technical lead: built the chatbot, experiment pipeline, and evaluation framework (team project)
- Built a retrieval-augmented chatbot answering Boston University Student Health Insurance Plan (SHIP) coverage questions, combining a FAISS vector index over a 12-document policy corpus (encoded with a sentence-transformers embedding model) with Qwen2.5-1.5B-Instruct for generation, and benchmarked it against retrieval-only and LLM-only baselines
- Designed a 30-prompt red-teaming set spanning normal, ambiguous, and adversarial coverage questions, and ran a 240-run experimental grid across the three system modes, two temperatures, and two retrieval depths, triaging every response into a five-way decision label (covered, not covered, prior authorization required, escalate, or abstain)
- Labeled every response for correctness, hallucination, and harm severity, finding the RAG and LLM-only modes correct on only 27.5% and 16.7% of prompts respectively versus 100% for retrieval-only, and that incorrect answers carried an average harm severity of about 4.1 out of 5, meaning the chatbot was often confidently wrong on higher-stakes insurance questions

Freight Plus: Optimizing Freight Logistics Through Analytics 2025
Suffolk University, ISOM 330 (Applied Statistics and Predictive Analytics). High-value goods segment lead (team project)
- Led the high-value goods segment of a team engagement analyzing shipment data for a Boston-area logistics firm under NDA, evaluating how factors including mileage, weight, ship type, delivery type, and time to cover drive freight pricing
- Built and pruned a classification decision tree to 3 terminal nodes via cross-validation for the high-value goods segment in R, and iterated between log and square-root transformations of price in a parallel linear regression model to satisfy model-fit assumptions, evaluating both against a held-out test split

Case Study: The Lego Discovery Center Boston's Quest for Growth 2023
Suffolk University, SBS-101 (Business Foundations)
- Worked with a team to develop strategic solutions for the Lego Discovery Center in Boston covering membership retention, attendance patterns, and competitive positioning, analyzing industry benchmarks and competitor offerings to recommend ways to increase annual pass sales and strengthen renewal rates

SKILLS

Languages and Tools: SQL (SQL Server, SSMS), Python, Power BI, Tableau, Excel, BigQuery (Google Cloud), R, Git, SAP, Streamlit, Vercel, Supabase, TypeScript, SQLite, Claude Code (AI agents), automated testing

Python Libraries: Pandas, NumPy, Scikit-learn, XGBoost, SHAP, NetworkX, Matplotlib, Seaborn, Hugging Face, LightGBM, pyfixest, GeoPandas, FAISS, sentence-transformers

Analytics Methods: Funnel analysis, A/B testing and causal inference, regression and panel models, classification, clustering and PCA, time-ordered model validation, model explainability (SHAP), LLM evaluation and red-teaming

Web and App Development: React, Next.js, Vite, Tailwind CSS, FastAPI, Docker, Cloudflare Workers and D1, REST API design, authentication (JWT, Cloudflare Access), shell and PowerShell scripting, scheduled jobs, schema design and migrations

Data Sources and Research: WRDS (I/B/E/S, Compustat, CRSP), FRED, UN COMTRADE, Qualtrics, Jupyter

Other: MS Office, Adobe Creative Cloud (Premiere Pro, Photoshop)
Spoken Languages: English (Native/Bilingual), Russian (Native/Bilingual), Azerbaijani (Proficient), Turkish (Proficient), Spanish (Intermediate)